

EXPERIMENT 15:

AIM:

To implement probabilistic CFG parsing.

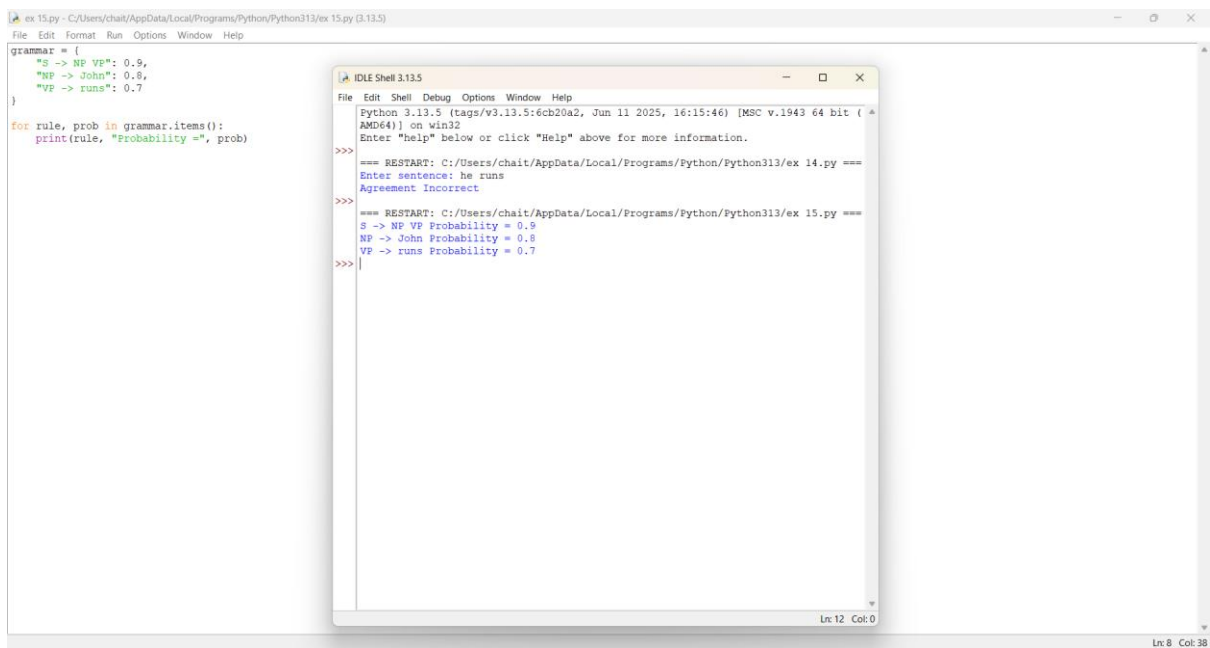
PROGRAM:

```
grammar = {  
    "S -> NP VP": 0.9,  
    "NP -> John": 0.8,  
    "VP -> runs": 0.7  
}
```

for rule, prob in grammar.items():

 print(rule, "Probability =", prob)

OUTPUT:



```
ex 15.py - C:/Users/chaht/AppData/Local/Programs/Python/Python313/ex 15.py (3.13.5)  
File Edit Shell Debug Options Window Help  
grammar = {  
    "S -> NP VP": 0.9,  
    "NP -> John": 0.8,  
    "VP -> runs": 0.7  
}  
for rule, prob in grammar.items():  
    print(rule, "Probability =", prob)  
  
IDLE Shell 3.13.5  
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>> == RESTART: C:/Users/chaht/AppData/Local/Programs/Python/Python313/ex 14.py ==  
Enter sentence: he runs  
Agreement Incorrect  
>>> == RESTART: C:/Users/chaht/AppData/Local/Programs/Python/Python313/ex 15.py ==  
S -> NP VP Probability = 0.9  
NP -> John Probability = 0.8  
VP -> runs Probability = 0.7  
>>> |  
Ln 12 Col: 0  
Ln 8 Col: 38
```